

Review

Explainability and Interpretability in Electric Load Forecasting Using Machine Learning Techniques – A Review

Lukas Baur ^{a,b,*}, Konstantin Ditschuneit ^c, Maximilian Schambach ^c, Can Kaymakci ^{a,b}, Thomas Wollmann ^c, Alexander Sauer ^{a,b}

^a Fraunhofer Institute for Manufacturing Engineering and Automation IPA, Nobelstraße 12, Stuttgart, 70569, Germany

^b Institute for Energy Efficiency in Production (EEP), University of Stuttgart, Nobelstraße 12, Stuttgart, 70569, Germany

^c Merantix Momentum GmbH, Max-Urich-Straße 3, Berlin, 13355, Germany

HIGHLIGHTS

- Publications on explainable and interpretable models increased at above-average rate.
- Interpretable models are either classical, a decomposition, fuzzy or probabilistic.
- Explainable methods for load forecasting include feature importance and attention.
- Open XAI research is standardization, interpretation of attention and causality.

ARTICLE INFO

Keywords:

Electric load forecasting
Explainability
Interpretability
Structured review

ABSTRACT

Electric Load Forecasting (ELF) is the central instrument for planning and controlling demand response programs, electricity trading, and consumption optimization. Due to the increasing automation of these processes, meaningful and transparent forecasts become more and more important. Still, at the same time, the complexity of the used machine learning models and architectures increases.

Because there is an increasing interest in interpretable and explainable load forecasting methods, this work conducts a literature review to present already applied approaches regarding explainability and interpretability for load forecasts using Machine Learning. Based on extensive literature research covering eight publication portals, recurring modeling approaches, trends, and modeling techniques are identified and clustered by properties to achieve more interpretable and explainable load forecasts.

The results on interpretability show an increase in the use of probabilistic models, methods for time series decomposition and the use of fuzzy logic in addition to classically interpretable models. Dominant explainable approaches are Feature Importance and Attention mechanisms. The discussion shows that a lot of knowledge from the related field of time series forecasting still needs to be adapted to the problems in ELF. Compared to other applications of explainable and interpretable methods such as clustering, there are currently relatively few research results, but with an increasing trend.

1. Introduction

Electric Load Forecasting (ELF) is a crucial component of modern electrical power systems, as it is relevant because of its application in power allocation-, energy trading- and demand-response optimization, and power system- and maintenance-planning, as well as for capacity sizing [1,2]. An ever-increasing installation of renewables increase the importance of accurately predicting future energy demands to schedule residual energy generation and balance energy network loads. In addition to forecasting loads at grid level, forecasting plays an important role in the industrial context, which accounts for a large proportion of the demand for storage. Large industrial consumers can

have substantial advantages if they participate directly in the energy market and improve their energy management by creating microgrids or by shifting loads over time. Here, the quality of the forecast has a positive effect on operations [3].

Machine and deep learning models have recently gained popularity in modeling ELF due to their ability to handle large amounts of data, better abstraction through automatic feature extraction, easier automation, and good performance [4,5]. However, these deep learning models often lack interpretability. That is, they act as black boxes and do not offer access to or understanding of the underlying decision-making process. This lack of interpretability and explainability can

* Corresponding author at: Fraunhofer Institute for Manufacturing Engineering and Automation IPA, Nobelstraße 12, Stuttgart, 70569, Germany.

E-mail address: lukas.baur@ipa.fraunhofer.de (L. Baur).

Acronyms

ML	Machine Learning
AI	Artificial Intelligence
ELF	Electric Load Forecasting
PLF	Probabilistic Load Forecasting
HLF	Hierarchical Load Forecasting
AQR	Additive Quantile Regression
GAM	Generalised Additive Models
SVD	Singular Value Decomposition
DMD	Dynamic Mode Decomposition
GPR	Gaussian Process Regression
BSTS	Bayesian Structural Time Series Model
SVM	Support Vector Machine
SVR	Support Vector Regression
DEMD	Differential Empirical Mode Decomposition
EMD	Empirical Mode Decomposition
QPSO	Quantum-behaved Particle Swarm Optimization
AR	Autoregressive
FCM	Fuzzy Cognitive Maps
FST	Fourier Series Transformation
FTS	Fuzzy Time Series
XAI	Explainable Artificial Intelligence
SHAP	Shapely Additive Explanations
LIME	Local Interpretable Model-agnostic Explanations
LSTM	Long Short-Term Memory
RNN	Recurrent neural network
SSM	State-Space Model
CAM	Class Activation Function

result in several consequences for load forecast applications, such as reduced trust in the model, difficulty in identifying model errors, and ineffective decision-making, as intransparent models “cannot be entirely accepted and utilized” [6]. In addition, when load forecasts are used to detect anomalies, a user is often interested not only in a corresponding detection but also in the reason behind the deviation [7].

This paper provides an overview of interpretability and explainability methods applied to machine learning models in ELF applications and highlights current research trends. We review the existing literature in the field and give an overview of the different methods of improving the interpretability and explainability of their ELF models.

The remainder of the paper is structured as follows: Section 1 defines the terms interpretability and explainability (Section 1.1), differentiates the problem from the more general Time Series forecasting (Section 1.2), and presents related and prior work (Section 1.3). By analyzing these, a literature gap emerges that will be closed by this work. Section 2 explains the method used for conducting the review. Sections 3 and 4 discuss the results of the review. The work concludes with a discussion (Section 5) and summary (Section 6).

1.1. Terminology

The terms *explainability* and *interpretability*, as used in the remainder, need to be defined first as there is no general agreement in the Machine Learning (ML) community, as [8] point out.

Rudin [9] argues that there is no single definition of interpretable ML, but there are “several forms of interpretable models”. Interpretability and explainability are closely related [8], and it can be difficult to distinguish between them. However, we can provide definitions

that help to understand the conceptual differences relevant to ELF applications.

In the context of this review, we find that many approaches to ELF explicitly consider how humans will interpret the model’s prediction. This leads us to choose the following definition of interpretability, which is focused on how humans can understand the model:

Definition 1 (Interpretability). Interpretability is the extent to which a human can understand a model’s prediction based only on the model’s design without additional information.

For example, a standard linear model is considered interpretable, as the direct contribution of each individual feature to the output target can be fully understood based on the feature’s coefficient. In this case, the coefficients are not a mere explanation of the output but a complete understanding of the model’s decision-making. Similarly, Decision Trees are considered interpretable under the above definition, as the output can be fully understood based on the decision boundaries at each node.

Here, with *design* the definition includes everything that is necessary for understanding the prediction of the model, i.e. in addition to the model architecture, possibly also necessary (data pipeline) pre-processing steps. Examples of this are data (dimensionality-) reduction and feature extraction: An interpretable model applied to non-traceable transformed (black-box) input features does not significantly increase the overall interpretability. However, design does not include the way in which the model determines the parameters during training, as the specific choice of parameters is irrelevant for understanding the predictions.

The above definition includes more advanced model classes like fuzzy logic and additive models, which are relatively easy for humans to understand. However, it excludes more complex concepts like random forests and attention maps, which can be difficult to reason about for humans. This example makes it clear that interpretability, by definition, is not a binary property but spans a spectrum. Also we agree with [7] that the interpretability of a model or an architecture depends to a certain degree on subjective expert and domain knowledge. For example, random forests are sometimes considered interpretable because they can be viewed as merely a collection of decision trees. However, we believe that the large number of decision trees typically used in random forests makes it difficult for humans to understand the overall prediction.

This distinction between *reasoning* about a prediction and *understanding* it motivates our definition of explainability.

Definition 2 (Explainability). Explainability is the extent to which a human can reason about the factors influencing a model’s prediction.

For example, input- and output relations can be calculated after a model has been trained to investigate the influence of input variables, as well as the model robustness. Still, this information does not give insights into how the model makes its predictions nor how it will perform on unseen data. Similarly, attention maps can be used to visualize the importance of individual components that lead to a prediction. However, the individual components and underlying architectures related to the attention map can still be high-dimensional, complete, and difficult for humans to understand.

Note that this definition does not restrict explainability to investigating black box models. However, unlike interpretability, explainability is achieved through post-hoc analysis of a trained model rather than a model’s inherent design. Again, explainability is a spectrum rather than a binary property. We will see that, with the help of these definitions, we can cluster the existing literature into both categories, with most ML approaches lying exclusively in one category.

Explainability and Explainable Artificial Intelligence (XAI) will be used interchangeably in this work.

1.2. ELF characteristics and delimitation

Load forecasting is a special case of the more general time series forecasting. While the latter predicts domain-independent time series e.g. financial, weather or sales data, the focus of the ELF is on predicting future electrical loads and modeling suitable input variables. A good overview of common use cases, fault dimensions and models can be found in [10,11].

These are typical forecasting horizons which are divided into the (Very) Short-Term, Medium-Term and Long-Term forecast, with time ranges up to 1 h, up to seven days, up to one year and beyond, respectively [10]. In addition to the cross-horizon integration of historical load variables (hourly, daily, weekly, etc.) and calendar information (special events, holidays, maintenance planning, living and working behavior), typical additional variables are included in the forecast in relation to the horizons: weather variables (temperature, humidity, wind, rain, cloud cover) for Short- and Medium-term and economic indicators for Medium- and Long-Term (population size, growth) [12]. The pre-processing also includes manual feature engineering to encode domain knowledge [13], as good knowledge expertise is required [12].

Further, load data is typically available hierarchically, which can also be used to improve forecasts, as is being investigated in the Hierarchical Load Forecasting (HLF) sub-field of research [14,15].

Due to this large number of parameters, ELF has a significant level of uncertainty [16], especially for Short-Term forecasts which count for the largest share (80% [10]), while at the same time load profiles are difficult to predict due to the complexity of the variables, their non-linearity and multiple seasonalities [17–19]. All these characteristics motivate an in-depth literature analysis of existing work for interpretability and explainability that goes beyond that of general time series forecasting.

1.3. Related work and literature gap

An initial search for related review papers, hereafter referred to as *meta-review*, was conducted to delineate areas already pursued and researched.

This section will briefly present the results of this meta-review, the other published reviews' thematic focuses, and conclude with an identification of literature gaps. The latter will be resolved in upcoming Sections through an extensive literature search (see Sections 2–5).

To identify potential gaps, review papers were sought that addressed the topics of *machine learning*, *load forecasting*, and *explainability* or *interpretability* in terms of content. Table 1 shows the results of the meta-reviews. Matching review papers were rated according to content and focus using a Harvey ball notation. An empty Harvey ball states that the topic was not addressed, a full one that the topic was the focus of the investigation. A partially filled ball indicates a thematic cut, but the topic was not the focus of the investigation or was addressed only superficially.

The following three related reviews covered, at least partially, all three search areas:

- The review paper by Machlev et al. [23] focuses on the use of explainable Artificial Intelligence (AI) for energy and power systems by addressing XAI potentials and problems for individual energy applications such as demand side management, energy scheduling, non-intrusive load monitoring, and forecasting. However, load forecasting is only addressed superficially as one of many energy applications. Furthermore, no explicit distinction is made between interpretability and explainability; the load forecasting results are mainly explainable.
- The tutorial and literature survey by Benidis et al. [29] focuses on deep learning methods for time series forecasting, of which load forecasting is a subfield. Although aspects of interpretability and explainability are mentioned for the presented models, the paper focuses on a complete picture of the forecasting methods rather than the applied methods of explainability and interpretability.

- The review paper by Xu et al. [30] summarizes knowledge about the interpretability of machine learning in smart grid applications. In addition to fault diagnosis, security, and stability analysis, the use case energy forecasting is briefly touched upon but not elaborated in depth.

In conclusion, according to our preliminary investigation, no review paper could be identified that fully addresses the addition terms in focus, thus resulting in the literature gap this publication aims to fill. To the best of our knowledge, this is the first review paper to focus specifically on methods of explainability and interpretability for load forecasting.

2. Used method

Our method of investigation follows that of Kitchenham [31] and Webster and Watson [32] with minor modifications. Initially, preliminary research identifies a potential knowledge gap and elaborates relevant attributes to ensure the necessity of the review and to derive research questions. The research was prepared and conducted using a review protocol; the overall search and filter process is depicted in Fig. 1. The scope of this paper is the intersection of the three research areas *interpretability* and *explainability*, *load forecasting*, and *machine learning*. Hence, the following search query was stated

```
1 ["XAI" or "interpretable" or "interpretability"
   or "explainable" or "explainability"]
2 and "load forecasting"
3 and ["DeepML" or "Deep Learning" or "Machine
   Learning" or "AI" or "artificial
   intelligence"]
```

and potentially interesting paper candidates were sought: Due to their size and widespread use, the databases *Web of Science*, *Science Direct*, *IEEE Xplore*, *ACM Digital Library*, *Springer Link*, *Wiley Online Library*, *Scopus*, and *Semantic Scholar* were used for the search. Where supported, the search string was applied only to the title and abstract. For search engines that did not allow such a restriction, the full text was searched first, and then abstracts and titles were downloaded from all hits and automatically filtered afterwards. After merging all results and removing the duplicates, this initial search resulted in 52 hits.

The syntactic filtering was followed by a manual screening in which further publications were filtered out: five could not be accessed; for one, only the title and abstract were in English, but the content was in Chinese; one was not a scientific article. Screening the abstracts resulted in nine further exclusions, as six papers did not directly address load forecasting, only related problems and three did not address interpretability or explainability.

The remaining 36 articles were studied and read in detail. Fig. 2 shows these results based on the initial screening. The reference ELF trend (red) was received by aggregating the results from *Web of Science*, *IEEE Xplore* for the respective years, and the trend for interpretability and explainability in ELF (dashed blue) results from yearly aggregating the initial screening's results, exponentially smoothed.

In the final step, papers that either focused on interpretability or explainability and covered machine learning were retained in the context of load forecasting.

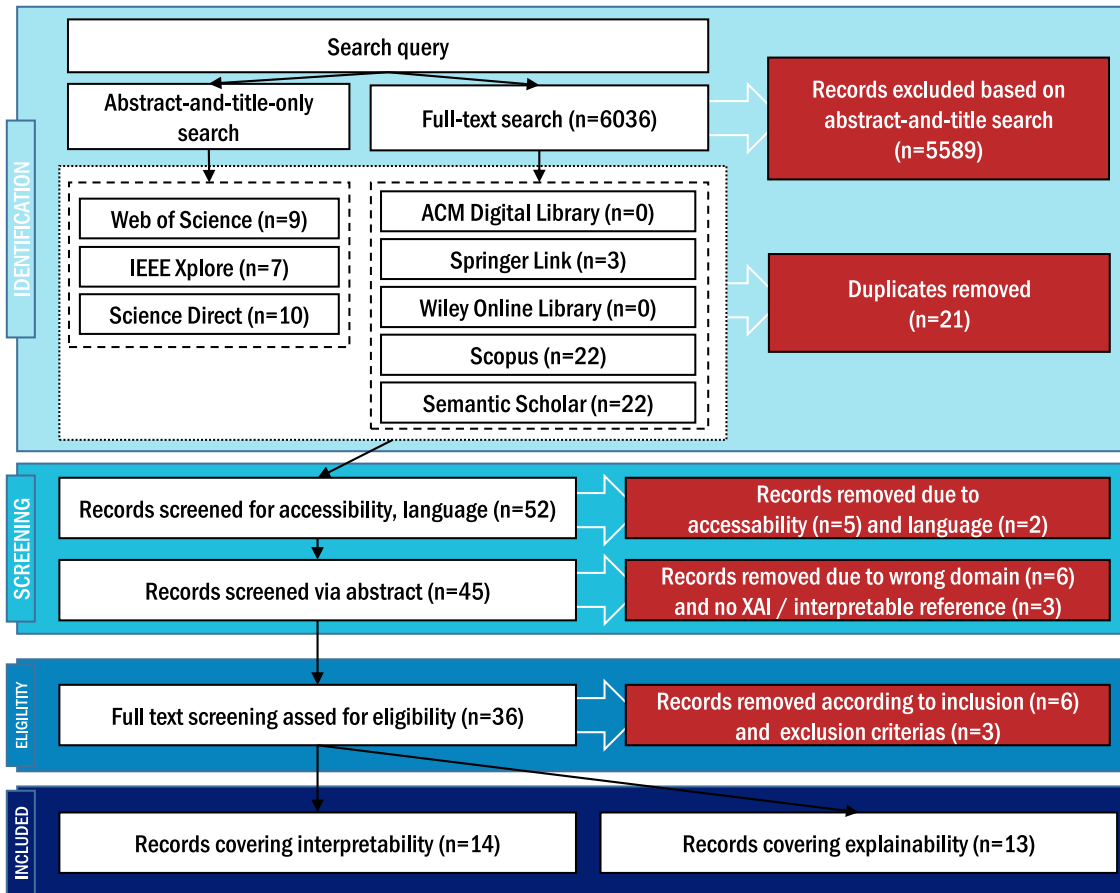
The inclusion criteria include papers that are thematically related to the original search and have a sufficient scientific level (peer-reviewed). As a result, five respectively one were excluded. The reasons for the three exclusions were: superficial description and no reproducibility (2 instances) and the assertion that the design was interpreted, but without proper justification or explanation. The articles were then clustered based on the definition (Section 1.1) into either interpretability or explainability and then grouped according to the model types or common approaches used.

This resulted in 14 articles on interpretability and 13 on explainability. The summary of the results is shown in Tables 2 and 3, respectively.

Table 1

Results of our meta-review, collecting previous review works on interpretable and/or explainable load forecasting.

Prior review	Covers		
	LF	I/E	Deep ML
“Explainable Artificial Intelligence (XAI) on TimeSeries Data: A Survey” by Rojat et al. [20]	○	●	○
“Time-series forecasting with deep learning: a survey” by Lim and Zohre [21]	○	○	●
“Artificial intelligence and machine learning approaches to energy demand-side response: A systematic review” by Antonopoulos et al. [22]	○	○	●
“Explainable Artificial Intelligence (XAI) techniques for energy and power systems: Review, challenges and opportunities” by Machlev et al. [23]	○	●	●
“Electricity load forecasting: a systematic review” by Nti et al. [10]	●	○	○
“Load forecasting techniques and methodologies: A review” by Singh et al. [24]	●	○	○
“A Review of Deep Learning Methods Applied on Load Forecasting” by Almalaq and Edward [25]	●	○	●
“A survey on deep learning methods for power load and renewable energy forecasting in smart microgrids” by Aslam et al. [26]	●	○	●
“Deep learning methods and applications for electrical power systems: A comprehensive review” by Ozcanli et al. [27]	●	○	●
“Electrical Load Forecasting Models for Different Generation Modalities: A Review” by Azeem et al. [28]	●	○	●
“Deep Learning for Time Series Forecasting: Tutorial and Literature Survey” by Benidis et al. [29]	●	○	●
“Review on Interpretable Machine Learning in Smart Grid” by Xu et al. [30]	●	●	○

**Fig. 1.** Schematic overview of the used literature search process (design idea based on [33]).

While the interpretable results can be categorized into the four clusters classical, time series decomposition, fuzzy logic and probabilistic load forecasting, explainable methods follow approaches to feature importance and attention. A detailed presentation and methodological classification of these results will be presented in the following two Sections, followed by a critical evaluation of missing model classes.

3. Interpretable LF methods

As a result of our comprehensive literature search and filtering, 14 works on interpretable load forecasting were found. All selected papers claim to be interpretable in their prediction process. Interpretability includes the understanding of a forecast on the model design and its

Table 2
Applied techniques to ensure interpretability based on the literature review.

Technique	Interpretability achieved through	Used in
Rule Based	Clear ramification, model inherently understandable	Moon et al. [6]
Regression	Clear transformation from input to output.	Mir et al. [34]
Autogressive models	Linear relationship between last values and predicted values	Esponzoza et al. [35]
Probabilistic models	Ensemble learning, inherited properties from deterministic models	Li et al. [36]
Probabilistic models	State space model, quantifying the posterior uncertainties	Roth et al. [37]
Probabilistic models	State space model, parameterized by a RNN	Arora et al. [38]
Time series decomposition	Split in simpler, meaningful time series to use models of lower complexities	Peng et al. [39], Fan et al. [40], Saengngam and Tonggumnead [41]
Probabilistic time series decomposition	For almost-periodic time series, splits trend and residual. The latter one is modeled with a stochastic process	Dylewsky et al. [42]
Fuzzy Logic	Modeling the time series as fuzzy time series and determining transition probabilities	de Lima e Silva et al. [43]
Fuzzy Logic	Fuzzy knowledge system with fewer rules or modeling special days	Ghanbari et al. [44], Feng and Liu [45]
Fuzzy Logic	Calculation of a Fuzzy Cognitive Map for understanding the interconnections	Orang et al. [46]

Table 3
Applied techniques to ensure explainability based on the literature review.

Technique	Explainability achieved through	Used in
Feature Importance (local)	Using LIME to calculate feature importance scores by locally fitting a linear model	Grzeszczyk and Grzeszczy [47]
Feature Importance (global)	Feature importance through mutual information to find correlations between input and output	Han et al. [48]
Feature Importance (local+global)	Using SHAP values explain the importance of features to individual or multiple predictions	Wu et al. [49], Wu et al. [50], Bolstad et al. [51], Li and Wang [1]
Feature Importance (local+global)	Using SHAP values in combination with coefficient if determination	Lee et al. [52]
Feature Importance (local+global)	ELF-specific method to estimate feature importance by learning to mask out unimportant features	Gürses-Tran et al. [53]
Feature Importance (local+global)	Compares SHAP values, and LIME scores with Permutation Importance (using ELI5)	Kuzlu et al. [54]
Attention mechanisms	Evaluating attention maps in combination with transformer or LSTM architectures	Gao et al. [55], Wang et al. [56], Li et al. [57], Xu et al. [58]

architecture. Therefore, in contrast to the methods of explainable forecasting, all presented approaches are model-specific, i.e., the techniques used refer to a compatible model type or a class of structurally similar models. Based on our findings, we sorted the results into the following four groups:

- The first group focuses on classical models which are inherently interpretable.
- The second group integrates the mechanism of time series decomposition to allow for simpler models.
- The third group focuses on different approaches related to and integrating Fuzzy Logic as a basis for decision-making.
- The last group discusses probabilistic forecasting and how it relates to the interpretability notation.

In the following, these clusters are discussed in more detail.

3.1. Classical interpretable models

To begin with, many model outputs can be easily interpreted if the model complexity is low. This concerns all models whose final computation, thus not their training, can be reduced to easily understandable operations. Besides classical regression decision trees, which base their prediction on simple branched if-then-else, linear regressors are easy-to-interpret models used in load forecasts. Regression models infer the unknown target load from existing relatively safe exogenous input variables.

Moon et al. [6], present a model based on Cubist and compare it with 12 other algorithms. The Cubist model, originally only a commercial product, is an extension of the classic decision tree. More specifically, it is an ensemble rule-based model tree but different from a regression tree, (1) different splitting criteria are used, (2) terminal node models are used instead of averages, and (3) typically, predictions arise from the combination of the terminal account model and

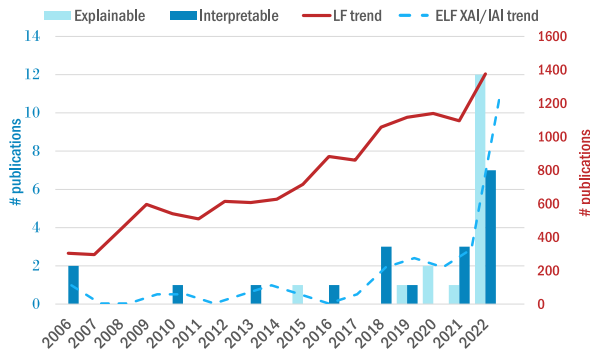


Fig. 2. Chronological classification according to the publication date of the found publications, grouped according to interpretability, explainability and their aggregated trend (blue, axis left) and the overall ELF research trend (red, axis right). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the search path models [59]. The model guarantees interpretability since it is possible to understand how the model makes individual decisions, and rules can then be derived from the tree. For further visualization, feature-important scores can be calculated for each forecast to determine the influence of the variables on the forecast, as [6] proposed.

Mir et al. [34] show that interpretability, model complexity, and model performance are not necessarily opposing properties of a predictive model: a linear regression prediction with a strong dependence on the exogenous variable yields good results if further polynomial combinations are added to the input variables in the context of feature engineering. The coefficients of the terms resulting from the training permit interpretable conclusions on the influence of the variables.

Another simple but widely used class of models are autoregressive models, i.e., models that make future predictions by a weighted linear combination of past data points and integrated exogenous variables. To evaluate the tradeoff between interpretability and model performance, [35] compare three autoregressive models with exogenous input with three parameterizations: In the simplest case (linear parameterization), the coefficients correspond precisely to the variable weights. In the second case (black box parameterization), coefficients are trained that weight an unknown nonlinear feature map (box function), similar to the kernel of a Support Vector Machine (SVM). In the last case (Partially Linear Parameterization), weather and calendar information are weighted nonlinearly, while historical values are weighted linearly. The case study for forecasting in a Short-Term load forecasting setting shows that the partial linear variant with partially interpretable coefficients performs just as well as the fully black-box-based variant.

3.2. Interpretability through time series decomposition

A recurring pattern as an aid to interpretability is the decomposition of the time series to be forecast into individual partial forecasts, which are then again combined into one global forecast.

The decomposition serves several purposes in terms of interpretability. Firstly, the decomposition reduces the forecast complexity of the decomposed signals so that partly less complex models, each of which can be interpreted separately, can be used. Second, the decomposed functions, such as *Trend* and *Season*, are based on semantics that the user can inherently understand. The three steps of decomposition, sub-forecasting, and aggregation are the essential components of this procedure, as visualized in Fig. 3.

The work of Peng et al. [39] combines simple concepts, each of which can be interpreted individually, to obtain a powerful yet interpretable hybrid model. The load is decomposed into a set of

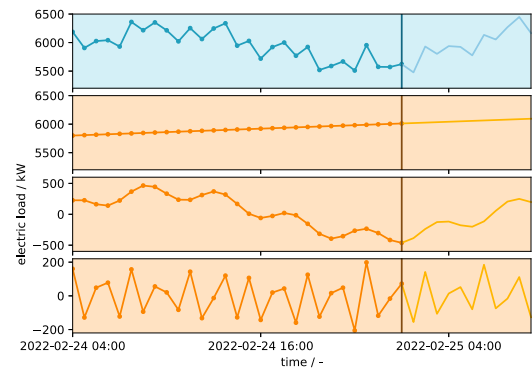


Fig. 3. Example of a time series decomposition for the sake of interpretability. The original signal (blue, dotted) is decomposed into sub-signals (orange, dotted), which are then forecasted separately (orange, solid) and aggregated to a final prediction (blue, solid). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

high-frequency signals, and a low-frequency residual is done using an Empirical Mode Decomposition (EMD) method (namely Differential Empirical Mode Decomposition (DEMD)). Then, the sub-forecasts are calculated separately on the two data sets: The prediction of the residual signal data is obtained using an Autoregressive (AR) model. The prediction on the high-frequency data is performed using Support Vector Regression (SVR). The SVR hyperparameters are tuned using Quantum-behaved Particle Swarm Optimization (QPSO). Summing the forecasts yields the aggregated forecast.

Fan et al. [40] also use EMD for decomposition to split the energy load profile into high-frequency curves and trend components. The sub-forecasts for high-frequency signal and trend are determined by a SVM and an AR model, respectively. The empirical analysis of the New South Wales dataset shows that the hybrid model performs better than the direct modeling using SVM. The authors conclude from the results that interpretability and model performance are not mutually exclusive.

In the work of Saengngam and Tonggumnead [41], two decomposition strategies based on Fourier Series Transformation (FST) for separating Long-Term and midterm energy consumption were modeled in a seasonal-trend setting. One model extends the FST by extracting additive, linear and multiplicative trends, then determining periodic seasons and recombining the signal based on these. The other model extends the first by adding further constraints to obtain a smoother trend. Regarding the results, in direct comparison with the AR models ARIMA and SARMIA, the two presented models showed better results, are easier to learn by parameter estimation, and are well interpretable.

Dylewsky et al. [42] propose a modeling approach for time series that are almost periodic based on Dynamic Mode Decomposition (DMD). Interpretability results from modeling the quasi-periodic trend and stochastic influences separately. Therefore, the lagged-values matrix is first decomposed using Singular Value Decomposition (SVD) and mapped to a lower-rank representation for calculating the periodic trend using DMD. The residual between the trend term and the actual load is called force and is modeled by a stochastic process using Gaussian Process Regression (GPR). For the forecast, the linear trend and several samples of the GPR are sampled in an ensemble fashion to obtain a probabilistic forecast. Since peaks in the calculated force values indicate strong deviations from the trend, further insights are possible.

3.3. Interpretability through Fuzzy logic

At the core of fuzzy logic, values are assigned to specific sets (*fuzzy sets*) using a *membership function*. Unlike in classical set theory, values can be partially assigned to sets, and typically, the membership is

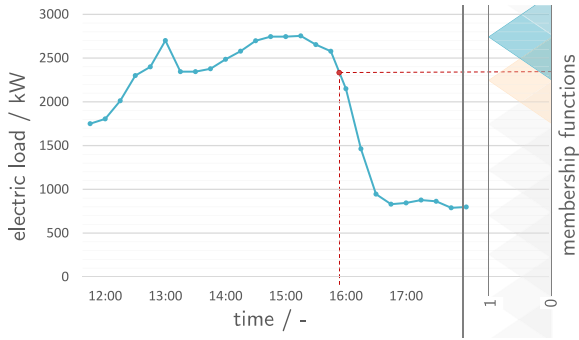


Fig. 4. Example to illustrate FTS as an interpretable method. Each moment (red) of load signal (blue) can be assigned to respective membership functions μ_i . Two membership functions (orange and blue) are non-negative in this example. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

defined on the interval $[0, 1]$, indicating a range between *no membership* and *fully included*. Based on these fuzzy sets, rules can be derived and functions defined [60].

A central concept proposed by Song and Chissom [61] in this context is the Fuzzy Time Series (FTS) forecasting. In the concept of FTS, the codomain of the time series is divided into buckets, and the membership is expressed through a fuzzy membership function as visualized in Fig. 4 on a fictional example. The FTS forecast then determines the transition probabilities between these memberships. In [43], this concept is extended by a simple integration for multivariate data. Based on the results of the experiments, this approach shows good accuracy, versatility, and computational performance despite its simplicity.

Although a fuzzy logic-based prediction system is based on comprehensible modeling, the prediction can become very complex due to a potentially high number of rules. The work of Ghanbari et al. [44] extends a genetic approach for learning a fuzzy knowledge base of an expert system through clustering, resulting in a significantly lower number of fuzzy rules with good performance compared to other fuzzy approaches. This shows once again that complex approaches comply with easy-to-interpret approaches.

One advantage of fuzzy logic systems is that they require relatively little data for training. This is particularly helpful for modeling atypical days, e.g., public or consecutive holidays, or bridging days, i.e., days preceding or following holidays, of which there are typically only a few in a training dataset. In [45], the authors focus on improving the load forecast for these anomaly days based on fuzzy rules, as, on the one hand, the load behavior deviates significantly on these days, and, on the other hand, little data is available, as the proportion of typical days to atypical days is very unbalanced. This procedure ensures that the few abnormal days can be modeled and interpreted separately.

Fuzzy Cognitive Maps (FCM) is another class of interpretable methods for time series and load forecasting consisting of a digraph that expresses the relationship of nodes (concepts) using weighted directed edges. The direction and weights between concepts model the effect of one node on another, as Fig. 5 explains. It is a knowledge-based model for forecasting by extracting knowledge of a fuzzy time series. Overall, FCMs are trained in two stages: First, the structure has to be learned using strategies like clustering or predefining a certain granularity. Then, the structure weights have to be learned using a weight matrix. In their work, Orang et al. [46] present a random-based method to reduce the computational cost of training such a graph model.

3.4. Interpretability through probabilistic modeling

A repeatedly cited drawback of interpretable load forecasting algorithms is that uncertainties are neither sufficiently quantified nor

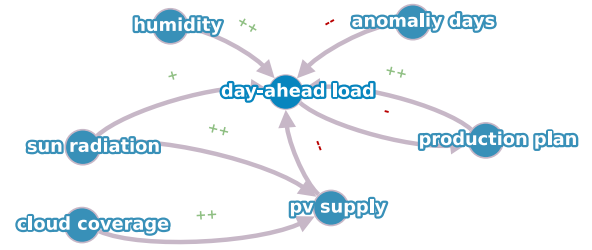


Fig. 5. Example to illustrate FCM.

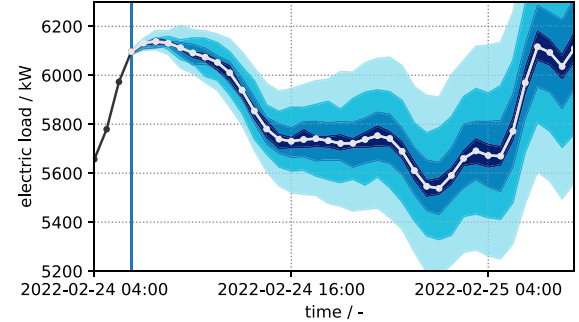


Fig. 6. Example of a probabilistic load forecast that, in addition to the most likely guess (white), shows the calculated quantile levels for the forecast (graduated in blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

made visible to the end user. Probabilistic Load Forecasting (PLF) aims to quantify and integrate uncertainties into the forecast. This can be done by estimating quantiles on the predictions or some level of additional confidence boundaries. Fig. 6 shows a visualization example of a probabilistic prediction by plotting selected quantiles.

We recognize that the class of probabilistic models cannot be fully classified as interpretable as defined in Section 1.1, and we see PLF at the far edge of the interpretability spectrum. While the additional modeling helps to better understand the prediction without requiring additional information for the user, the underlying models can be very complex, and the design depends on how the PLF is made.

For the classification of interpretability, first, the different modeling techniques are described. According to [62], three approaches for calculating a PLF can be distinguished in general: (1) deterministic models that become ensemble learners by passing in different input scenarios, (2) novel PLF models that implicitly learn a posterior probability by definition and (3) post-processing of a point forecast model by integrating the residual likelihood function or combining multiple models.

Models created according to approaches (1) and (3) inherit the interpretability properties of their deterministic models. The easy-to-interpret concept of a SVM, for example, was augmented by an ensemble-learning setup in the work done by Li et al. [36]. For this purpose, SVMs were trained on subsamples of the entire training set, and the hyperparameters of each model were determined using swarm optimization. The SVMs were then sampled using Monte Carlo simulation to obtain both final forecasts and confidence intervals.

Models following approach (2) internally model the probabilities, that yield inherent interpretability in the sense that likelihoods are modeled directly.

A popular base class are State-Space Model (SSM), i.e. classical statistical models that allow interpretable statements to be made about trends and seasonality by construction, which are then extended. [37], for example, achieve interpretability by using a Bayesian Structural Time Series Model (BSTS), which combines a SSM at its core with quantifying the posterior uncertainties of the model components. Since

all state-space components can be added independently, the model's understandability and interpretability increase. In addition to the classic PLF in the day-ahead setting, the integration of the submeter time series was investigated. For this purpose, improvements by adding different features were investigated by examining the model coefficients (covariances). Thus, relevant submeter time series could be distinguished from irrelevant ones, and their influences could be quantified. Arora et al. [38] present a hybrid model called *StateNeuron*, which combines the interpretability and probabilistic prediction of an SSM with the learning performance of a neural network. For this purpose, an innovation state-space model, which belongs to the class of exponential smoothing, is parameterized by an Recurrent neural network (RNN). They validated their model on the two known GEFCom-14 and ISO-NE datasets and compared it with 9 other models, including with and without external weather variables.

Although the work of Sigauke et al. [63] claims an interpretable modeling approach, it shows that the statistical models used (namely Generalised Additive Models (GAM), Additive Quantile Regression (AQR)) can become complex and are only interpretable to a very limited extent.

4. Explainable LF methods

The increasing demand for transparent AI systems has also driven a growing interest in the development of XAI methods. As they are designed in a post-hoc way, existing models can be extended so that users can reason about and better understand the load forecasts' influencing factors independent of the model's complexity. We have identified 14 previous studies that have employed XAI methods in ELF. These studies can be broadly categorized into two groups:

- The first group focuses on estimating the importance of features in model predictions.
- The second group uses attention mechanisms to reveal interdependencies between features.

While applying XAI methods in load forecasting is still in its early stages, these studies already provide valuable insights into the factors driving energy consumption and help to improve model accuracy and robustness. In the following, we provide an overview of each study's main findings and contributions in these two groups.

4.1. Explainability through feature importance

Several recent studies have used black box XAI methods to gain insights into the overall importance of individual features to the prediction of machine learning models. The importance of features can be computed for individual predictions, called *local feature importance*, aggregated over multiple predictions or even for the whole dataset, called *global feature importance*.

Ribeiro et al. [64] introduce Local Interpretable Model-agnostic Explanations (LIME), a common method to compute local feature importance. LIME calculates local feature importance scores by locally fitting an interpretable linear model to the complex, generally non-linear black box model of interest. That is, explainability is achieved by local interpretability. More formal, for a given sample $x \in X$ and a function $f: X \rightarrow \mathbb{R}$, LIME approximates f using a linear function $g: X \rightarrow \mathbb{R}, g(x) = w \cdot x$ at the sample x . It chooses the weights w to minimize the cost function $\sum_{z, z' \in \Pi(x)} (f(z) - g(z'))$, where $\Pi(x)$ is a given neighborhood around the sample. An ELF related example of this is given in Fig. 7.

Grzeszczyk and Grzeszczyk [47] use LIME to explain the influence of features on individual predictions in their Short-Term load forecasting system. In particular, they evaluate their model on an untypically cold day and find that the temperature is the most important feature for that prediction.

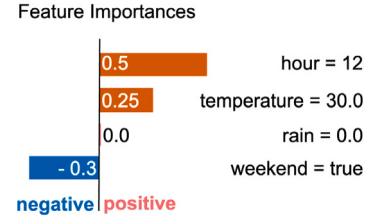


Fig. 7. The figure displays a typical bar plot of LIME values. LIME can compute feature importance values for a single data point at a time. These are usually color-coded depending on their sign and plotted using a bar chart, providing the user with an instant overview of a feature's impact. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

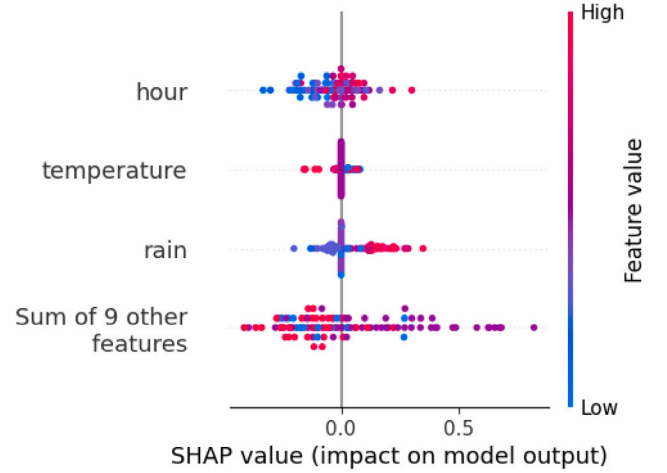


Fig. 8. Distribution of SHAP, values computed by [49] for a subset of their dataset with especially cold days. For each feature (here: *hour*, *temperature*, *rain*, and the other 9), the input values were scattered along the x-axis according to their impact on the model's prediction. The color of each scatter point indicates how large the input value was in relation to all others. The figure was created using the openly available SHAP-package by [65]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Lundberg and Lee [65] introduce another popular method to compute local as well as global feature importances called Shapely Additive Explanations (SHAP). SHAP values explain the importance of features to the predictions of a model for individual or multiple inputs. The exact computation of these SHAP values is complicated, and we refer the interested reader to the original paper.

Wu et al. [49] investigate their forecasting system based on a Long Short-Term Memory (LSTM) architecture using SHAP values, enabling them to capture the black box model's local and global properties. They find that the hour of the day has the largest influence on the prediction overall. However, they can only use SHAP values to gain insights into a subset of their data. Fig. 8 shows the SHAP values computed for winter days. SHAP's ability to marginalize individual features allows the authors to analyze their model's predictions for all winter days at once and to extract correlations and distributional properties. From the values in Fig. 8, it can be said that *temperature* has a negative and *rain* has a predominantly positive impact on the forecasting outcome, but mainly for large values in absolute terms. *Hour*'s impact is scattered more widely, indicating overall high importance, similar to the other nine variables.

Henriksen et al. [50] investigate their LSTM hour-ahead load forecasting model similarly. [51] propose a day-ahead load forecasting MLP model and evaluate its predictions using SHAP. Using the tool's ability to evaluate a subset of features, they investigate the impact of the public Easter holiday in Norway. As expected, the holiday has a large negative effect on the prediction.

[52] train an XGBoost model, a specific implementation of a Gradient Boosted Tree, for Short-Term load forecasting and introduce a new variant of SHAP values to analyze feature importances. They use the coefficient of determination to consider the local variance of SHAP values and reduce the effect of random noise on the computed importance values. [1] train an XGBoost model to predict the next day's load for a region in China. They find that their system mainly considers the previous day's load, but, perhaps surprisingly, whether the previous days were holidays also impacts the prediction. [48], on the other hand, train a random forest model but leverage mutual information [66] to find correlations between feature and target variables.

In addition to general methods in the context of XAI, such as LIME and SHAP, there are also some developments specific to ELF. Gürses-Tran et al. [53] introduce a custom XAI method for load forecasting that tries to estimate the importance of features by learning to mask out unimportant features. It has conceptual similarities to both LIME and SHAP but has not been widely used.

By employing these black box explainability methods, researchers identified key features that significantly impact load forecasting accuracy, such as temperature, humidity, and time of day. Furthermore, these methods have enabled the identification of potential biases in the model, such as over-reliance on specific input features or the presence of non-linear interactions between features. [54] summarize the application of black box XAI in their comparison of LIME, SHAP, and a third model-independent XAI library named ELI5 [67] in the context of load forecasting. ELI5 has “built-in support for several ML frameworks” and visually provides ways for explaining black box models. Kuzlu et al. [54] is making the following observations that align with the findings of the other works: While LIME and SHAP are both model-agnostic and can explain individual predictions, SHAP values additionally provide global insights into multiple predictions while providing theoretical guarantees, namely consistency and local accuracy. However, a “key limitation of all tools is that they need to run many evaluations of the original mode”. That is, they are usually computationally expensive. This restriction of the application to lightweight models might be why more expensive models, such as transformer-based load forecasting models, have not been investigated using these tools.

4.2. Explainability through attention

Load forecasting has witnessed an upsurge in deep learning models incorporating attention mechanisms, particularly due to the increased interest in attention-based transformer architecture [68] in recent years. The attention mechanism allows the model to selectively focus on relevant input features, as shown in Fig. 9. This mechanism allows the model to capture intricate relationships between input features and their effect on the output. Additionally, attention maps provide the explainability of the model's decisions, making it a critical aspect of XAI. As in the previous section, attention maps are employed for local explanations of individual predictions and global explanations of the model's general prediction patterns.

Gao et al. [55] train an additive model consisting of a seasonality component and a transformer model [68] to forecast the electrical load for the next day. Their transformer model thus has to predict the residual between the actual and seasonal loads. They observe that the attention matrices of most layers focus on features between 3 pm and 11 pm of the previous days the most. The authors “suspect that this is due to the higher uncertainty of people's activity” at these times. However, they also find structured attention maps that are hard to interpret, “indicating complex nonlinear mapping relationships”, according to the authors.

Wang et al. [56] approach multi-entity load forecasting using a transformer model that employs cross-attention to learn the implicit co-dependence amongst the load of multiple buildings. By aggregating

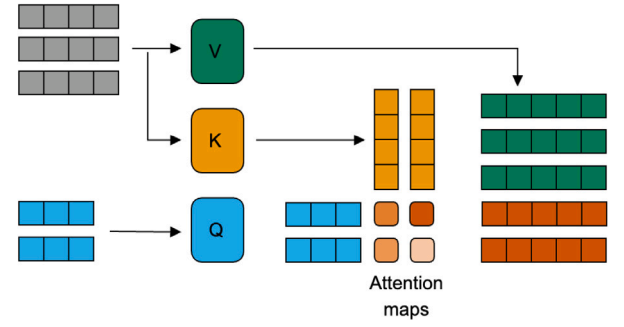


Fig. 9. Attention commonly refers to a mechanism that uses a learnable and input-dependent mask to construct weighted sums of feature maps. This is commonly used to learn relationships across sequential data. In this figure, the cross-attention mechanism is depicted. Given two sequences of feature vectors, the features are linearly projected to so-called Queries (Q), Keys (K), and Values (V). Attention is then calculated between Queries and Keys, for example, using the cosine similarity, and the obtained scalar attention values are used to weigh the corresponding Values. The assigned attention scores are then commonly used to provide insights into the computations of the models as they provide an approximate measure of the information flow across different inputs within the sequence.

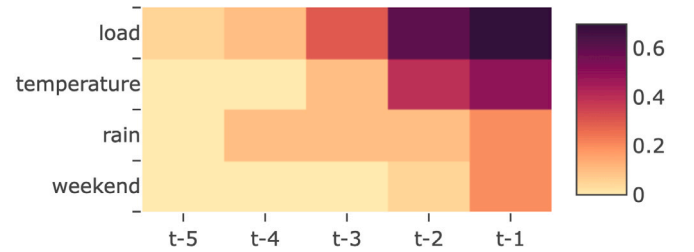


Fig. 10. This example of a resulting aggregated attention map depiction, inspired by [57], reveals the most common usage of attention maps to explain model predictions. For a single prediction, higher values in the attention maps indicate a higher weighting of the corresponding feature at that timestep. This weighting is commonly interpreted as a measure of importance.

the cross-building attention maps, they find that only a single building receives significant attention from other buildings. Which they find reasonable, as this building has “photovoltaic panels installed [and is] the only building with electricity generating capacity, and its energy consumption pattern differs” and thus provides additional information to the other buildings’ forecasts. On the other hand, the average self-attention maps to other lagged timesteps enable insights into temporal relationships learned by the model. They find that “the amount of attention the network can provide diminishes over time”. Similarly, [56,57] integrate variants of the attention mechanism into LSTM-based neural networks. They also aggregate attention maps to receive comparable insights. Fig. 10 shows an example attention map from [57].

Xu et al. [58] introduces LSTM-based attention mechanism, that allows for both multistep-predictions and probabilistic forecasting. In each prediction time step, the mixture attention mechanism was used “to capture different temporal dynamics in a multivariate sequence”. When tested on residential load data, two explainability aspects are particularly advantageous: In addition to local and global feature importance, effects on variable-wise temporal importance can also be analyzed and visualized, which helps users to better understand patterns and relationships.

While the attention mechanism has proven to be a powerful tool for improving the accuracy of electrical load forecasting models, we also see it as a valuable tool for explaining the inner workings of load forecasting models, often helping to improve the understanding of the data and enabling guided improvement of a model. But it has limitations: Attention is primarily used in specific deep learning architectures, such as transformers and LSTMs, and may not be applicable

to other architectures. As apparent from the above examples, there is currently no standardized way to interpret attention in deep learning models, making it difficult to compare the explanations provided by the different approaches.

Like the black box XAI methods, attention can only reveal correlations between features, but it does not provide insight into the underlying mechanisms that drive the model's predictions. As a result, attention-based explanations are insufficient to fully understand why a model makes its predictions.

5. Discussion

5.1. Current situation and trend

Load forecasts play an important role in research, as they are directly needed on the way to automated and decentralized company control. As more and more companies use load forecasting as a basis for planning critical processes, interpretable and explainable models are becoming increasingly important. This **trend** is shown in this research: While load forecasting has become increasingly popular, especially in the last five years, the number of publications on explainable and interpretable models has increased at an above-average rate (see Fig. 2).

Nevertheless, this study indicates that the relative share is still a **very small part** of the overall load forecasting research landscape. The authors hypothesize that this is due to two reasons. Firstly, the classical models developed for image classification or time series classification are difficult to transfer to regression problems with multivariate and strongly (auto)correlated input values, as is the case for the more general time series forecasting [21]. Furthermore, load forecasts are rarely used as a stand-alone solution, but are used to solve problems based on them, such as energy purchasing, loading and unloading management. An extensive list of applications that use a load forecast as input can be found in [24]. As a result, the explanation of the overall system is more interesting and decision-critical than the actual forecast.

Based on this work, it can be stated that the proportion of interpretable approaches to explainable ones is still **roughly balanced**. Nevertheless, the ratio of explainable methods has been increasing recently.

The results show that the research directions for interpretable load prediction models are widely spread. There is no abstract or recurring pattern that can be applied to all models; instead, the solutions are model-specific and, therefore, individual.

5.2. Critics and limitations

Interpretability and user's background

As already indicated above, complexity cannot always be measured objectively. While the range of classical methods is large (Section 3.1), the respective degree of complexity depends strongly on the personal understanding of the model and the model's size (number of submodels, parameters, modeling terms and constraints), so that it is difficult to give an objective measure of interpretability.

Inherited reliability and evaluation issues

There is a general debate about the **reliability** of general post-hoc methods and cases, in which explanations mislead [69] and whether attention can be used as a tool of explainability [70,71]. For example, there are a number of examples where either explanations can be manipulated by systematically changing the input [72,73] or problems with domain shifts occur.

Another problem with the use of pos-hoc methods is that they typically ignore the **sequential dependency** between inputs [21]. This must be taken into account, especially for the forecast of electrical loads where the inputs are auto-correlated and periodic (as with time lags)

on all forecast horizons [74–76]. This makes the use of these methods problematic for time series [21].

In addition to the reliability problems mentioned above, the general problems with the **evaluation** of model explainability are inherited from the abstract methods for time series prediction. A distinction is made between qualitative (expert judgment, domain knowledge) and quantitative (e.g. perturbation) evaluation [77]. Due to the complex inputs (feature vector with historical and exogenous features, see Section 1.2) and output (real-valued vector or vector of distributions is the case of PLF), a quantitative evaluation is preferable for ELF. For quantitative metrics, however, a ground truth typically has to be available [78], which makes the evaluation of explanations difficult overall and requires further methods [79].

Decomposition of load

Decomposing the forecasting task into individual sub-forecasts (Section 3.2) can help to extract more interpretable components, such as trends and seasonalities, which in themselves allow interpretable statements to be made. Nevertheless, it should be noted that the aggregated overall forecast can only be interpretable if this applies to all subforecasts. A **recursive use** of interpretable approaches is therefore necessary and thus limits this approach.

Probabilistic forecasting and interpretability

The results of the review show that the quantification of uncertainties plays a crucial role in the context of ELF. Since it is a cross-sectional technique that primarily contributes to the XAI target **confidence** [80] through probabilistic modeling, a clear classification in the XAI spectrum is difficult so that an exact undercutting of the PLF classes is necessary: In our opinion, the PLF construction using ensemble learners and variable input scenarios (Monte Carlo), says little about how the input variables transform to a prediction (interpretability) but describes the effects of the (potential black box) model and acts as a post-processing and analysis step. **Inherently probabilistic models**, however, are classified as interpretable models according to our definition as the models allow a more general statement by analyzing the learned distribution functions.

Regardless of the precise classification, it can be emphasized that probabilistic approaches provide helpful additional information [81] for a better understanding [82], and make an important contribution to the use for grid operators, retail companies, demand response aggregators, customers, and electricity market bidders [83]. There is also a need [84] to increase the interpretability of these models.

5.3. Open challenges and future work

While applying explainable AI methods in load forecasting has seen significant advancements, several avenues for future exploration remain.

Transfer from related research areas

Compared to other research disciplines (such as image classification) and the more general problem of time series forecasting, there are still some approaches that have not yet been used for ELF. This paragraph summarizes some of these approaches.

According to [21], there are two classes for post-hoc analysis. While many hits from the area of model approximation with methods such as LIME and SHAP were found (Section 4.1), **gradient-based** methods are underrepresented. Gradient-based methods analyze model gradients to determine which inputs mostly impact the loss functions [21]. Proposed representatives are **saliency maps** (as in [85,86]) and **influence functions**. The former was originally developed in form of Class Activation Functions (CAMs) for computer vision tasks to identify image parts relevant for classification [85], but can be conceptually transferred to time series forecasting [87]. For influence functions, the prediction of a model is traced back to the training data in order to identify the training

points that have led most to this prediction [88]. Since the inputs are less intuitive and more difficult for the end user to visualize compared to image classifications in the area of ELF, a meaningful transfer to electric load forecasting problems must be examined.

Further, **Counterfactual Explanation** is an XAI method that provides explanations for the classification by generating an existing input that is as similar as possible to an input (“nearest unlike neighbor” [89]), but which causes a change in the prediction of the classifier. This makes it possible to understand what minimum changes need to be made to the input in order to effect a conscious change in the model output. Although this method is already used in time series classification (as in [90–92]), there is not yet a complete extension for regression-based time series forecasting (partially in [93]), and none for ELF. Especially in connection with peak electric load forecasting methods in a multistep forecasting setting, a counterfactual explanation could be a useful tool.

Additionally, there is a need to uncover not just feature correlations but also the underlying causal mechanisms that drive predictions. None of the ELF-papers examine **causal relationships** or use similar techniques for variable selection, as is has been used in the more general case of time series forecasting. In [94], for example, the *Granger causality test*, which is “a statistical test used to estimate the bivariate causality between two time series” is used to select features. Similarly, interesting influencing variables could be evaluated in residential and industrial load forecasting, the influences of load consumptions are diverse and complex. Causal relationships are particularly pronounced in the integration of specific variables such as machine occupancy schedules or mandatory rest periods in the industrial sector and fixed break and vacation times in the public sector. The integration of physical constraints would also be conceivable.

Lastly, there is a list of **interpretable models**, which has not been applied to and evaluated on a ELF setting. One such as the class is GAMs, where ELF applications already provide interesting results [95,96]. In [97] an overview of the characteristics of inherently interpretable models is shown and state-of-the-art models are evaluated in this respect.

Ultimately, further developments in XAI will allow for more transparent and trustworthy AI systems, that are critical for wider acceptance and optimal performance in the energy sector.

Merging ELF with control-system

ELFs are very important to solve further **downstream tasks** [98], such as electricity purchase and sale [99], scheduling of demand-side management such as flexibility and battery charging planning [100–103] and consumption anomaly detection [104]. In these models, the forecast serves as input for further optimization or decision-making. It is unclear whether and how the techniques for explainability and interpretability can be transferred to control systems that combine the two components by means of reinforcement learning, as in recent work [105]. A transfer of the reliable XAI techniques is crucial [106].

Hierarchical load forecasting

With the rapid expansion of smart meter installations in recent years [107] and methods that allow **hierarchical forecasts** to be linked in an improved way [108], new use cases for explainable and interpretable methods in the context of HLF could also emerge. HLF is an approach to calculate forecasts for hierarchical load time series by optimizing the performance of the model and the hierarchical structure [15]. It is unclear how the known methods (see also Section 3.2) can be transferred to multi-level structures and to what extent explainability and interpretability can be achieved in such an overall model.

Attention issues

Another critical challenge lies in the **standardization** and deeper interpretation of attention mechanisms in deep learning models, which will allow for more consistent comparisons and insights across studies [109]. With the use of different model architectures and layers, knowledge cannot be transferred across the board and interpreted specifically for each ELF application.

Given the increased relevance of interpretable and explainable models, especially due to the increased use of autonomous energy management based on forecasting, growth in this research field is expected. Here, approaches for other areas’ explainable and interpretable regression problems can and must be transferred.

6. Conclusion

Electric Load Forecasting (ELF) is a crucial component of modern electrical power systems, as many downstream applications rely on it. With increasing automation, the need for transparent forecast methods is growing. This paper reviews the current state of research on interpretability and explainability in the field of electric load forecasting using machine learning. To the best of our knowledge, this is the first structured review focusing on the electric load use case. For this purpose, eight online publication media sources were searched. The amount of literature on explainable and interpretable methods is about the same, but the former is methodologically more diverse due to its dependence on architecture and model design.

Due to our findings, **approaches to increase interpretability** can be either realized by classical inherently interpretable load forecasting models such as linear models, decision trees, additive or autoregressive models or in combination with time series decomposition. In addition, posterior uncertainties can be inherently understood in probabilistic models. Fuzzy logic to supplement classical binary decision-making systems receives new use cases in the context of interpretable load forecasting.

The current approaches to **increase explainability** of electric load forecasting methods particularly focus on feature importance-based techniques such as SHAP and LIME and attention mechanisms.

Compared to the more general research field of time series forecasting, there are similar **challenges and limitations**: there are relatively few explainable methods for (electric) time series forecasting compared to methods for (electric) time series classification, and evaluating these is challenging due to the complex structure of time series data in input and output. Furthermore, many concepts, although known for time series forecasting, have not yet been intensively tested for ELF, including gradient-based methods (such as Class Activation Functions (CAMs), or influence functions), Counterfactual Explanation and methods for investigating or finding causal relationships. It is unclear how these concepts will be extended to hierarchical load forecasting in smart grids and how the interpretability and explainability of ELF-based models will develop with the increase in reinforcement learning approaches.

CRedit authorship contribution statement

Lukas Baur: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Konstantin Ditschuneit:** Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis. **Maximilian Schambach:** Writing – review & editing, Supervision. **Can Kaymakci:** Writing – review & editing, Supervision. **Thomas Wollmann:** Supervision. **Alexander Sauer:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

We would like to thank the anonymous reviewers for their helpful insights and valuable additions. Further, the authors are thankful for the helpful feedback from Dr. Timm Kuhlmann and Dr. Christian Schneider, and the linguistic revision by Dr. Birgit Spaeth.

Funding

This work was supported by the German Federal Ministry of Economic Affairs and Climate Action (BMWK) through the project “FlexGUIDe” (grant number 03EI6065D).

References

- [1] Li M, Wang Y. Power load forecasting and interpretable models based on GS_XGBoost and SHAP. *J Phys Conf Ser* 2022;2195(1):012028.
- [2] Zhu J, Dong H, Zheng W, Li S, Huang Y, Xi L. Review and prospect of data-driven techniques for load forecasting in integrated energy systems. *Appl Energy* 2022;321:119269.
- [3] Bracale A, Caramia P, De Falco P, Hong T. A multivariate approach to probabilistic industrial load forecasting. *Electr Power Syst Res* 2020;187:106430.
- [4] Almalaq A, Zhang JJ. Deep learning application: Load forecasting in big data of smart grids. *Deep learning: Algorithms Appl* 2020;103–28.
- [5] Zhang L, Wen J, Li Y, Chen J, Ye Y, Fu Y, et al. A review of machine learning in building load prediction. *Appl Energy* 2021;285:116452.
- [6] Moon J, Park S, Rho S, Hwang E. Interpretable short-term electrical load forecasting scheme using cubist. *Comput Intell Neurosci* 2022;2022.
- [7] Petch J, Di S, Nelson W. Opening the black box: The promise and limitations of explainable machine learning in cardiology. *Can J Cardiol* 2022;38(2):204–13, Focus issue: New digital technologies in cardiology.
- [8] Marcinkevičs R, Vogt JE. Interpretability and explainability: A machine learning zoo mini-tour. 2020, arXiv preprint arXiv:2012.01805.
- [9] Rudin C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nat Mach Intell* 2019;1(5):206–15.
- [10] Nti IK, Teimeh M, Nyarko-Boateng O, Adekoya AF. Electricity load forecasting: a systematic review. *J Electr Syst Inf Technol* 2020;7(1):13.
- [11] Kuster C, Rezgui Y, Mourshed M. Electrical load forecasting models: A critical systematic review. *Sustain Cities Soc* 2017;35:257–70.
- [12] Mustapha M, Mustafa MW, Khalid SN, Abubakar I, Shareef H. Classification of electricity load forecasting based on the factors influencing the load consumption and methods used: An overview. In: 2015 IEEE conference on energy conversion. IEEE; 2015, p. 442–7. <http://dx.doi.org/10.1109/CENCON.2015.7409585>.
- [13] Voss M, Bender-Saebelkamp C, Albayrak S. Residential short-term load forecasting using convolutional neural networks. In: 2018 IEEE international conference on communications, control, and computing technologies for smart grids (smartGridComm). 2018, p. 1–6.
- [14] Hong T, Xie J, Black J. Global energy forecasting competition 2017: Hierarchical probabilistic load forecasting. *Int J Forecast* 2019;35(4):1389–99.
- [15] Zhang C, Li R. A novel closed-loop clustering algorithm for hierarchical load forecasting. *IEEE Trans Smart Grid* 2020;12(1):432–41.
- [16] Mandara CK, Jadoun VK, Agarwal A. Load forecasting of electric power distribution system using different techniques: A review. In: 2022 IEEE students conference on engineering and systems. 2022, p. 01–6.
- [17] Ain Q, Iqbal S. Enhancing short-term load forecasting using ES-dRNN with context vector. In: 2022 24th international multitopic conference. 2022, p. 1–6.
- [18] Hagan M, Behr S. The time series approach to short term load forecasting. *IEEE Trans Power Syst* 1987;2:785–91.
- [19] Abdoli G, Mehrara M, Ardalani ME. Forecasting IRAN's max daily demand for electricity in different weather types using deep learning approach. *Int J Eng Technol* 2019;11:198–206.
- [20] Rojati T, Puget R, Filliat D, Del Ser J, Gelin R, Díaz-Rodríguez N. Explainable Artificial Intelligence (XAI) on TimeSeries data: A survey. 2021, <http://dx.doi.org/10.48550/arXiv.2104.00950>, arXiv:2104.00950 [cs].
- [21] Lim B, Zohren S. Time-series forecasting with deep learning: a survey. *Phil Trans R Soc A* 2021;379(2194):20200209, Publisher: Royal Society.
- [22] Antonopoulos I, Robu V, Couraud B, Kirli D, Norbu S, Kiprakis A, et al. Artificial intelligence and machine learning approaches to energy demand-side response: A systematic review. *Renew Sustain Energy Rev* 2020;130:109899.
- [23] Machlev R, Heistrene L, Perl M, Levy KY, Belikov J, Mannor S, et al. Explainable Artificial Intelligence (XAI) techniques for energy and power systems: Review, challenges and opportunities. *Energy AI* 2022;100169, Publisher: Elsevier.
- [24] Singh AK, Ibraheem, Khatoon S, Muazzam M, Chaturvedi DK. Load forecasting techniques and methodologies: A review. In: 2012 2nd international conference on power, control and embedded systems. Allahabad, Uttar Pradesh, India: IEEE; 2012, p. 1–10. <http://dx.doi.org/10.1109/ICPES.2012.6508132>, URL: <http://ieeexplore.ieee.org/document/6508132/>.
- [25] Almalaq A, Edwards G. A review of deep learning methods applied on load forecasting. In: 2017 16th IEEE international conference on machine learning and applications. 2017, p. 511–6. <http://dx.doi.org/10.1109/ICMLA.2017.0-110>.
- [26] Aslam S, Herodotou H, Mohsin SM, Javaid N, Ashraf N, Aslam S. A survey on deep learning methods for power load and renewable energy forecasting in smart microgrids. *Renew Sustain Energy Rev* 2021;144:110992.
- [27] Ozcanli AK, Yaprakdal F, Baysal M. Deep learning methods and applications for electrical power systems: A comprehensive review. *Int J Energy Res* 2020;44(9):7136–57.
- [28] Azeem A, Ismail I, Jameel SM, Harindran VR. Electrical load forecasting models for different generation modalities: A review. *IEEE Access* 2021;9:142239–63, Conference Name: IEEE Access.
- [29] Benidis K, Rangapuram SS, Flunkert V, Wang Y, Maddix D, Turkmen C, et al. Deep learning for time series forecasting: Tutorial and literature survey. *ACM Comput Surv* 2022;55(6):1–36.
- [30] Xu C, Liao Z, Li C, Zhou X, Xie R. Review on interpretable machine learning in smart grid. *Energies* 2022;15(12):4427, Publisher: MDPI.
- [31] Kitchenham B. Kitchenham, B.: Guidelines for performing systematic literature reviews in software engineering. EBSE Technical Report EBSE-2007-01, 2007.
- [32] Webster J, Watson RT. Analyzing the past to prepare for the future: Writing a literature review. *MIS Q* 2002;xiii–xiii.
- [33] Liberati A, Altman DG, Tetzlaff J, Mulrow C, Gøtzsche PC, Ioannidis JP, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. *Ann Intern Med* 2009;151(4):W–65.
- [34] Mir AA, Ullah K, Khan ZA, Bashir F, Khan TUR, Altamimi A. Short term load forecasting for electric power utilities: A generalized regression approach using polynomials and cross-terms. *Eng Proc* 2021;12(1):21.
- [35] Espinoza M, Suykens JA, De Moor B. Structured kernel based modeling: An exploration in short term load forecasting. Technical Report, Technical Report 05-206, Leuven, Belgium: ESAT-SISTA, KU Leuven; 2006.
- [36] Li Y, Che J, Yang Y. Subsampled support vector regression ensemble for short term electric load forecasting. *Energy* 2018;164:160–70.
- [37] Roth J, Chadlawada J, Jain RK, Miller C. Uncertainty matters: Bayesian probabilistic forecasting for residential smart meter prediction, segmentation, and behavioral measurement and verification. *Energies* 2021;14(5):1481.
- [38] Arora P, Khosravi A, Panigrahi B, Suganthan P. Remodelling state-space prediction with deep neural networks for probabilistic load forecasting. *IEEE Trans Emerg Top Comput Intell* 2021;6(3):628–37.
- [39] Peng LL, Fan GF, Huang ML, Hong WC. Hybridized DEMD and quantum PSO with SVR in electric load forecasting. *Energies* 2016;9(3):221.
- [40] Fan GF, Qing S, Wang H, Hong WC, Li HJ. Support vector regression model based on empirical mode decomposition and auto regression for electric load forecasting. *Energies* 2013;6(4):1887–901.
- [41] Saengngam N, Tonggumnead U. Predicting Thailand electricity load demand with modified Fourier series and seasonal-trend decomposition methods using LOESS transformation. *Appl Sci Eng Prog* 2023;16(2):5720.
- [42] Dylewsky D, Barajas-Solano D, Ma T, Tartakovsky AM, Kutz JN. Stochastically forced ensemble dynamic mode decomposition for forecasting and analysis of near-periodic systems. *IEEE Access* 2022;10:33440–8.
- [43] de Lima e Silva PC, Severiano Jr CA, Alves MA, Cohen MW, Guimarães FG. A new granular approach for multivariate forecasting. In: Computational neuroscience: Second Latin American workshop, LAWCON 2019, São João Del-Rei, Brazil, September 18–20, 2019, Proceedings. Springer; 2019, p. 41–58.
- [44] Ghanbari A, Ghaderi SF, Azadeh MA. A clustering based genetic fuzzy expert system for electrical energy demand prediction. In: 2010 2nd international conference on computer and automation engineering, vol. 5, IEEE; 2010, p. 407–11.
- [45] Feng L, Liu Z. Effects of multi-objective genetic rule selection on short-term load forecasting for anomalous days. In: 2006 IEEE power engineering society general meeting. IEEE; 2006, p. 7–pp.
- [46] Orang O, Silva PCdL, Guimarães FG. Introducing randomized high order fuzzy cognitive maps as reservoir computing models: A case study in solar energy and load forecasting. 2022, arXiv preprint arXiv:2201.02158.
- [47] Grzeszczyk TA, Grzeszczyk MK. Justifying short-term load forecasts obtained with the use of neural models. *Energies* 2022;15(5):1852.
- [48] Han Y, Sha X, Grover-Silva E, Michiardi P. On the impact of socio-economic factors on power load forecasting. In: 2014 IEEE international conference on big data (Big Data). 2014, p. 742–7.

- [49] Wu K, Gu J, Meng L, Wen H, Ma J. An explainable framework for load forecasting of a regional integrated energy system based on coupled features and multi-task learning. *Prot Control Mod Power Syst* 2022;7(1):24.
- [50] Henriksen E, Halden U, Kuzlu M, Cali U. Electrical load forecasting utilizing an Explainable Artificial Intelligence (XAI) tool on Norwegian residential buildings. In: 2022 international conference on smart energy systems and technologies. 2022, p. 1–6.
- [51] Bolstad DA, Cali U, Kuzlu M, Halden U. Day-ahead load forecasting using explainable artificial intelligence. In: 2022 IEEE power & energy society innovative smart grid technologies conference. 2022, p. 1–5.
- [52] Lee YG, Oh JY, Kim D, Kim G. SHAP value-based feature importance analysis for short-term load forecasting. *J Electr Eng Technol* 2023;18(1):579–88.
- [53] Gürses Tran G, Körner TA, Monti A. Introducing explainability in sequence-to-sequence learning for short-term load forecasting. *Electr Power Syst Res* 2022;212:108366.
- [54] Kuzlu M, Cali U, Sharma V, Güler Ö. Gaining insight into solar photovoltaic power generation forecasting utilizing explainable artificial intelligence tools. *IEEE Access* 2020;8:187814–23.
- [55] Gao J, Hu W, Zhang D, Chen Y. TgDfL2.0: Theory-guided deep-learning for electrical load forecasting via transformer and transfer learning. 2022, arXiv, arXiv:2210.02448.
- [56] Wang Z, Zhu Z, Xiao G, Bai B, Zhang Y. A transformer-based multi-entity load forecasting method for integrated energy systems. *Front Energy Res* 2022;10:952420.
- [57] Li C, Dong Z, Ding L, Petersen H, Qiu Z, Chen G, et al. Interpretable memristive LSTM network design for probabilistic residential load forecasting. *IEEE Trans Circuits Syst I Regul Pap* 2022;69(6):2297–310.
- [58] Xu C, Li C, Zhou X. Interpretable LSTM based on mixture attention mechanism for multi-step residential load forecasting. *Electronics* 2022;11(14):2189.
- [59] Quinlan JR, et al. Learning with continuous classes. In: 5th Australian joint conference on artificial intelligence, vol. 92, World Scientific; 1992, p. 343–8.
- [60] Zadeh LA. Fuzzy logic. *Computer* 1988;21(4):83–93.
- [61] Song Q, Chissom BS. Fuzzy time series and its models. *Fuzzy Sets and Systems* 1993;54(3):269–77.
- [62] Hong T, Fan S. Probabilistic electric load forecasting: A tutorial review. *Int J Forecast* 2016;32(3):914–38.
- [63] Sigauke C, Nemukula M, Maposa D. Probabilistic hourly load forecasting using additive quantile regression models. *Energies* 2018;11(9).
- [64] Ribeiro M, Singh S, Guestrin C. “Why should I trust you?”: Explaining the predictions of any classifier. In: Proceedings of the 2016 conference of the North American chapter of the association for computational linguistics: demonstrations. 2016, p. 97–101.
- [65] Lundberg SM, Lee SI. A unified approach to interpreting model predictions. In: Guyon I, Luxburg UV, Bengio S, Wallach H, Fergus R, Vishwanathan S, et al., editors. *Advances in neural information processing systems* 30. Curran Associates, Inc.; 2017, p. 4765–74, URL: <http://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.pdf>.
- [66] Bishop CM, Nasrabadi NM. In: *Pattern recognition and machine learning*, vol. 4, Springer; 2006.
- [67] Korobov M, Lopuhin K. eli5. 2020, URL: <https://github.com/eli5-org/eli5>.
- [68] Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need. *Adv Neural Inf Process Syst* 2017;30.
- [69] Lipton ZC. The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery.. *Queue* 2018;16(3):31–57.
- [70] Jain S, Wallace BC. Attention is not explanation. 2019, arXiv preprint arXiv:1902.10186.
- [71] Wiegrefe S, Pinter Y. Attention is not not explanation. 2019, arXiv preprint arXiv:1908.04626.
- [72] Ghorbani A, Abid A, Zou J. Interpretation of neural networks is fragile. In: Proceedings of the AAAI conference on artificial intelligence, vol. 33, (01):2019, p. 3681–8.
- [73] Dombrowski AK, Alber M, Anders C, Ackermann M, Müller KR, Kessel P. Explanations can be manipulated and geometry is to blame. *Adv Neural Inf Process Syst* 2019;32.
- [74] Sood R, Koprinska I, Agelidis VG. Electricity load forecasting based on autocorrelation analysis. In: The 2010 international joint conference on neural networks. IEEE; 2010, p. 1–8.
- [75] Yadav V, Srinivasan D. Autocorrelation based weighing strategy for short-term load forecasting with the self-organizing map. In: 2010 the 2nd international conference on computer and automation engineering, vol. 1, IEEE; 2010, p. 186–92.
- [76] Bouktif S, Fiaz A, Ouni A, Serhani MA. Single and multi-sequence deep learning models for short and medium term electric load forecasting. *Energies* 2019;12(1):149.
- [77] Zhao Z, Shi Y, Wu S, Yang F, Song W, Liu N. Interpretation of time-series deep models: A survey. 2023, arXiv preprint arXiv:2305.14582.
- [78] Swamy V, Frej J, Käser T. The future of human-centric eXplainable Artificial Intelligence (XAI) is not post-hoc explanations. 2023, arXiv preprint arXiv:2307.00364.
- [79] Schlegel U, Arnout H, El-Assady M, Oelke D, Keim DA. Towards a rigorous evaluation of XAI methods on time series. In: 2019 IEEE/CVF international conference on computer vision workshop. IEEE; 2019, p. 4197–201.
- [80] Arrieta AB, Díaz-Rodríguez N, Del Ser J, Bannetot A, Tabik S, Barbado A, et al. Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Inf Fusion* 2020;58:82–115.
- [81] Liu B, Nowotarski J, Hong T, Weron R. Probabilistic load forecasting via quantile regression averaging on sister forecasts. *IEEE Trans Smart Grid* 2015;8(2):730–7.
- [82] Yang Y, Li W, Gulliver TA, Li S. Bayesian deep learning-based probabilistic load forecasting in smart grids. *IEEE Trans Ind Inf* 2019;16(7):4703–13.
- [83] Shepero M, Van Der Meer D, Munkhammar J, Widén J. Residential probabilistic load forecasting: A method using Gaussian process designed for electric load data. *Appl Energy* 2018;218:159–72.
- [84] Xu C, Sun Y, Du A, Gao D-c. Quantile regression based probabilistic forecasting of renewable energy generation and building electrical load: A state of the art review. *J Build Eng* 2023;107772.
- [85] Saadallah A, Jakobs M, Morik K. Explainable online deep neural network selection using adaptive saliency maps for time series forecasting. In: Joint European conference on machine learning and knowledge discovery in databases. Springer; 2021, p. 404–20.
- [86] Pan Q, Hu W, Zhu J. Series saliency: Temporal interpretation for multivariate time series forecasting. 2020, arXiv preprint arXiv:2012.09324.
- [87] Van Zyl C, Ye X, Naidoo R. Harnessing explainable artificial intelligence for feature selection in time series energy forecasting: A comparative analysis of Grad-CAM and SHAP. *Appl Energy* 2024;353:122079.
- [88] Koh PW, Liang P. Understanding black-box predictions via influence functions. In: International conference on machine learning. PMLR; 2017, p. 1885–94.
- [89] Schoenborn JM, Weber RO, Aha DW, Cassens J, Althoff KD. Explainable case-based reasoning: A survey. In: AAAI-21 workshop proceedings. 2021.
- [90] Delaney E, Greene D, Keane MT. Instance-based counterfactual explanations for time series classification. In: International conference on case-based reasoning. Springer; 2021, p. 32–47.
- [91] Karlsson I, Rebane J, Papapetrou P, Gionis A. Explainable time series tweaking via irreversible and reversible temporal transformations. In: 2018 IEEE international conference on data mining. IEEE; 2018, p. 207–16.
- [92] Ates E, Aksar B, Leung VJ, Coskun AK. Counterfactual explanations for multivariate time series. In: 2021 international conference on applied artificial intelligence. IEEE; 2021, p. 1–8.
- [93] Hao J, Shi Q, Ye Y, Zeng W. TimeTuner: Diagnosing time representations for time-series forecasting with counterfactual explanations. *IEEE Trans Vis Comput Graphics* 2023.
- [94] Hmamouche Y, Casali A, Lakhal L. A causality based feature selection approach for multivariate time series forecasting. In: DBKDA 2017, The ninth international conference on advances in databases, knowledge, and data applications. 2017.
- [95] Pierrot A, Goude Y. Short-term electricity load forecasting with generalized additive models. In: Proceedings of ISAP power, vol. 2011, 2011.
- [96] Yang L, Ren R, Gu X, Sun L. Interactive generalized additive model and its applications in electric load forecasting. In: Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining. 2023, p. 5393–403.
- [97] Sudjianto A, Zhang A. Designing inherently interpretable machine learning models. 2021, arXiv preprint arXiv:2111.01743.
- [98] Hahn H, Meyer-Nieberg S, Pickl S. Electric load forecasting methods: Tools for decision making. *European J Oper Res* 2009;199(3):902–7.
- [99] Xu W, Zhang P, Wen D. Decision-making model of electricity purchasing for electricity retailers based on conditional value-atRisk in day-ahead market. In: 2020 12th IEEE PES Asia-Pacific power and energy engineering conference. IEEE; 2020, p. 1–5.
- [100] Garulli A, Paoletti S, Vicino A. Models and techniques for electric load forecasting in the presence of demand response. *IEEE Trans Control Syst Technol* 2014;23(3):1087–97.
- [101] Abbas F, Feng D, Habib S, Rasool A, Numan M. An improved optimal forecasting algorithm for comprehensive electric vehicle charging allocation. *Energy Technol* 2019;7(10):1900436.
- [102] Pallonetto F, Jin C, Mangina E. Forecast electricity demand in commercial building with machine learning models to enable demand response programs. *Energy AI* 2022;7:100121.
- [103] Brinkel N, Visser L, van Sark W, AlSkaif T. A novel forecasting approach to schedule aggregated electric vehicle charging. *Energy AI* 2023;14:100297.
- [104] Wang X, Ahn S-H. Real-time prediction and anomaly detection of electrical load in a residential community. *Appl Energy* 2020;259:114145.
- [105] Specht JM, Madlener R. Deep reinforcement learning for the optimized operation of large amounts of distributed renewable energy assets. *Energy AI* 2023;11:100215.

- [106] Dey S, Marzullo T, Zhang X, Henze G. Reinforcement learning building control approach harnessing imitation learning. *Energy AI* 2023;14:100255.
- [107] Sovacool BK, Hook A, Sareen S, Geels FW. Global sustainability, innovation and governance dynamics of national smart electricity meter transitions. *Global Environ Change* 2021;68:102272.
- [108] Jiang Z, Wu H, Zhu B, Gu W, Zhu Y, Song Y, et al. A bottom-up method for probabilistic short-term load forecasting based on medium voltage load patterns. *IEEE Access* 2021;9:76551–63.
- [109] Chen X. The advance of deep learning and attention mechanism. In: 2022 international conference on electronics and devices, computational science. IEEE; 2022, p. 318–21.